

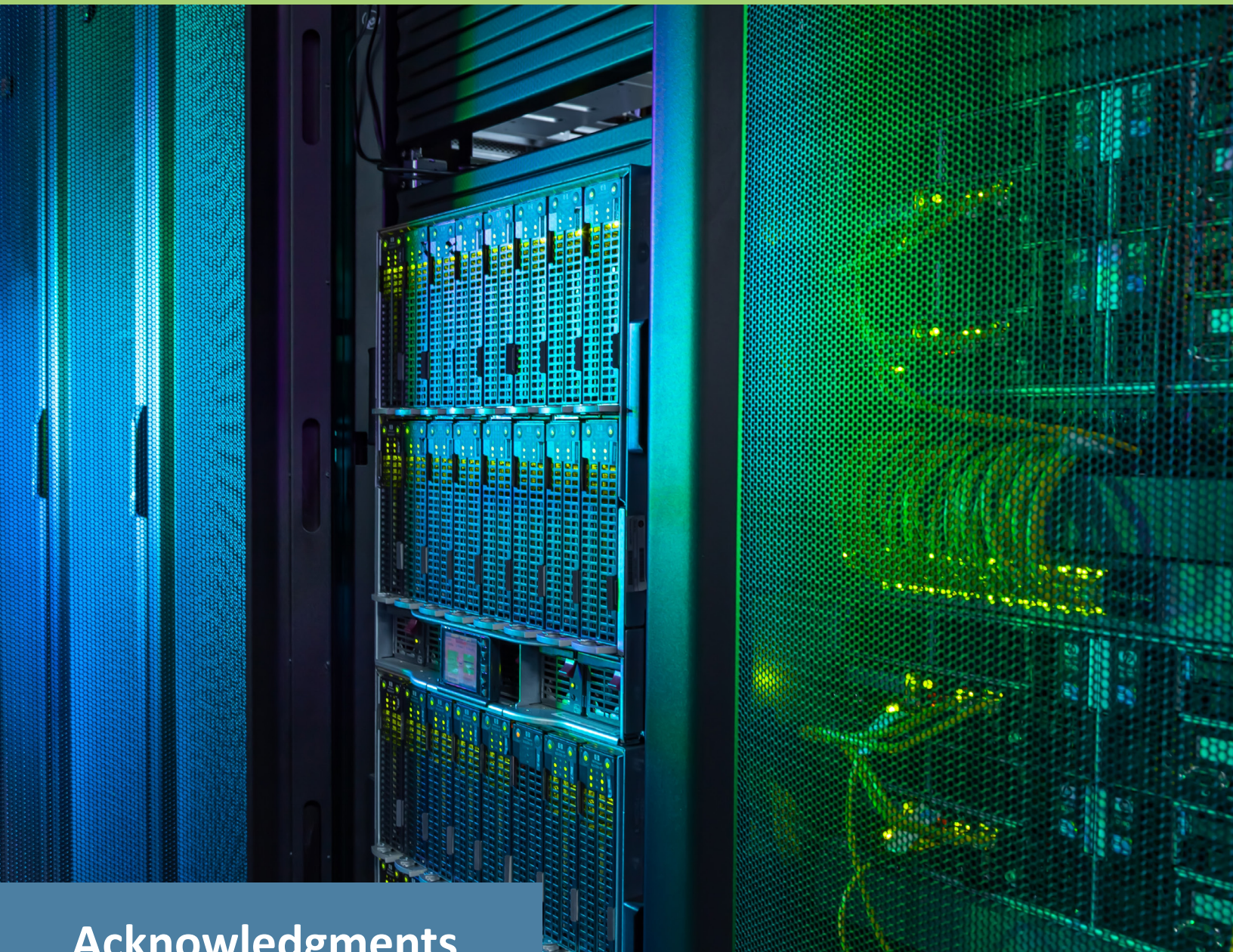


HARC

Thirsty Data and the Lone Star State: The Impact of Data Center Growth on Texas' Water Supply

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Acknowledgments

HARC is an independent, non-partisan non-profit research, policy and program implementation organization focused on sustainability solutions. Learn more at www.HARCresearch.org

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Executive Summary

If you ask state and local leaders across the country for their take on the AI and data center boom, they'll likely talk about the jobs and revenue this new industry could bring to their community. Ask them about the boom's impact on electricity and water use, and their excitement will fade quickly.

Everyone, it seems, is aware we are on the cusp of the most resource-hungry economic boom in modern history. And everyone wants to win the economic race.

No one, however, knows how we will meet the industry's insatiable demand for water and electricity.

Data centers are notoriously power hungry; one facility can require as much electricity as an entire town. And they're thirsty, too. Not only do they impact water demand through the amount of electricity they consume (electricity generation

requires a significant amount of water), but they also require water for on-site operation, specifically for cooling the acres of whirring servers and processors.

In the last year, the projected spike in data center electricity demand has received significant attention. Their water demand has only recently entered the mainstream discussion. Because water and electricity are inextricably linked, this paper discusses both and provides estimates of how this new industry could impact Texas' water supply, our communities, other businesses, and the 30 million Texans who already face significant water supply challenges.

It also includes some broad recommendations that state and local leaders should consider as more and more technology companies seek to connect to our grid and water systems.



Texas Data Centers at a Glance

As of September 2025, Texas has more than 400 data center facilities concentrated in five regions.¹

- Dallas / Fort Worth (197)
- Houston (48)
- San Antonio (60)
- Austin (53)
- West Texas (59)

Many more data centers are under construction or in planning and development phases.

Data Center Impact on Energy and Water Demand

- Collectively, existing data centers require approximately 9,567 megawatts (MW) of electricity.²
- Between existing facilities and forecasted growth, ERCOT estimates data centers could require 22,175 to 77,965 MW of capacity on the grid by 2030 – a nearly tenfold increase in only five years.³
- Between direct (on-site cooling) and indirect (electricity generation) water consumption, HARC estimates that existing data centers in Texas consume approximately 25 billion gallons (Bgal) of water (77 thousand acre-feet (kAF)) or 0.4% of total water use in 2025.
- Nationally, water use for cooling is expected to increase by 2-3x by 2028.⁴
- If data center growth projections hold, this demand could increase to 29-161 Bgal (98-494 kAF) by 2030, 0.5-2.7% of total annual water use in Texas.

Implications For a Thirsty State

- Texas' State Water Plan does not include projected demand growth for data centers. Because there are already unmet needs,

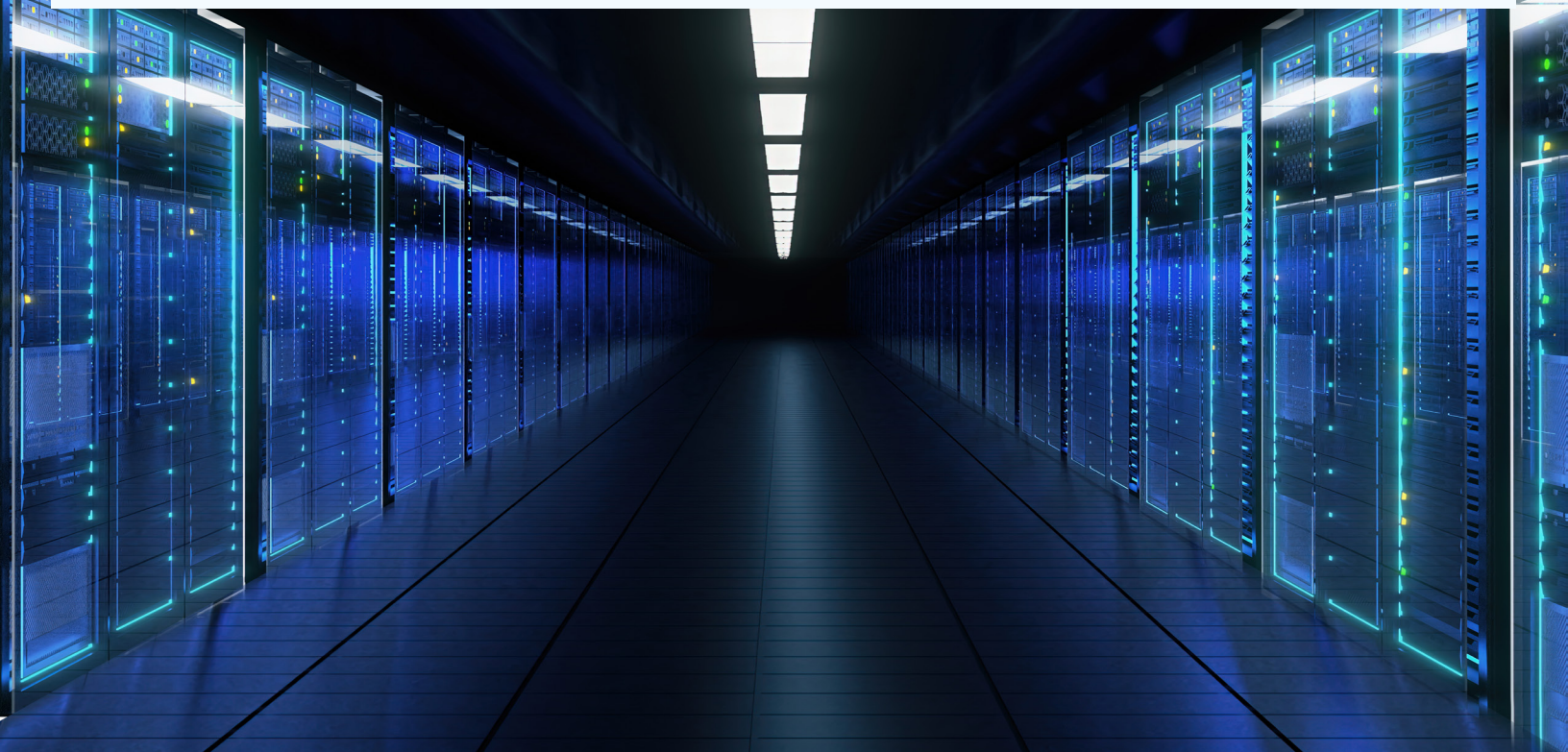
including the current 4.8-million-acre-foot shortage as determined by the Texas Water Development Board, data centers' unknown unmet needs are poised to place unprecedented stress on local water supplies across the state.

- Lack of water planning information impacts communities' capacity to request funding and shortens the time they have to develop new water supplies.
- Texas does not have guidelines or requirements to address large water demands through efficiency, conservation, or alternative water supplies.
- Location matters. Water stress in particular parts of the state will impact communities and other industries, including agriculture, differently.



Recommendations

- **Require Better Electricity and Water Demand Data** – Texas should require and release better electricity and water demand data from large users. Today, critical data center information, including the amount of water and energy they use, their emissions, and their community impacts, is significantly limited. Without better access to data for these facilities and other new major uses, state and local leaders' ability to anticipate and respond to their impact is significantly hindered.
- **Include Large Water Demand Uses in Water Planning** – The state should amend its process for forecasting demand in the State Water Plan to include potential large demands such as data centers.
- **Reform Groundwater Regulations** – State leaders should champion comprehensive groundwater regulation reform to better and more realistically align water demands and property rights with science-based scarcity projections.
- **Incentivize or Require Water Reuse or Conservation** – Local or state water and energy planners should consider incentivizing or requiring data center companies to deploy strategies to reduce water demands, including dry cooling, water lean energy sources (solar or wind, for example), energy efficiency, or alternative water supplies (reuse, brackish water, or oil and gas wastewater, for example).
- **Offset Water Use through Other Water-Saving Activities** – Save water in the same river basin or aquifer to offset the data center water use.
- **Leverage Infrastructure Investment** – Privately capitalize infrastructure with the data center as the primary customer and repurposed for public use later as the facility's needs decrease or be constructed large enough to support both needs.
- **Seek Public-Private Solutions** – Water providers, including cities, counties, and municipal utility districts, should seek public-private partnerships to develop new water supplies.



Thirsty Data and the Lone Star State: The Impact of Data Center Growth on Texas' Water Supply

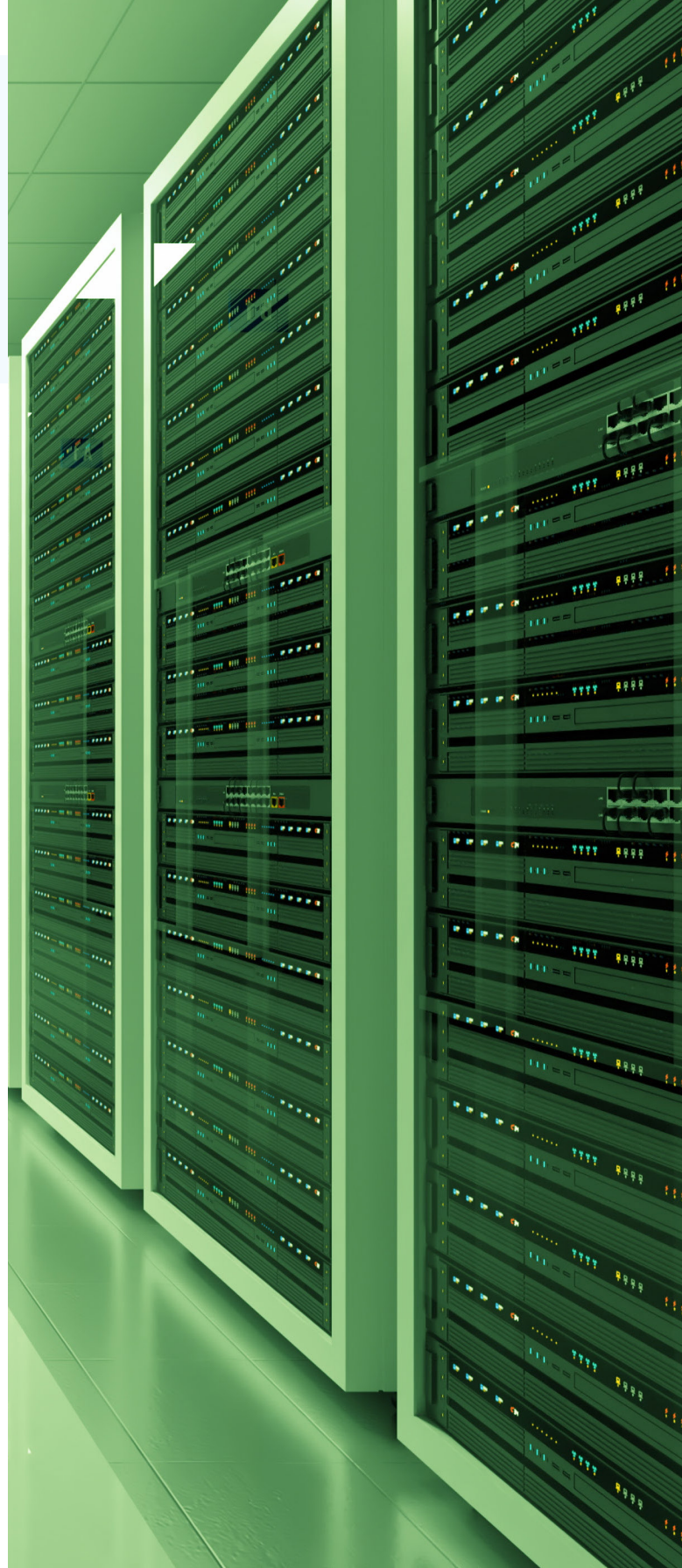
Recurring droughts, rapid population growth, and rising demand are already straining Texas' water resources, and data centers, a key component of the state's thriving digital economy, are among the fastest-growing water users.

Water plays a crucial role in the operation of data centers – indirectly, to produce the electricity that powers data center operations, and directly, to cool the servers and other equipment.⁵ Nationally, water use for cooling alone is expected to increase by 2-3x by 2028.⁶

Historically, data centers disproportionately use water resources from water-stressed watersheds.⁷ However, water usage by data centers varies, and issues with transparency have made determinations of water usage difficult.⁸ In Texas, where, as of April 2025, over 80% of the state is experiencing drought and 17% is experiencing exceptional drought, this consumption makes an even more substantial difference for local communities with limited water supplies.⁹

In this paper, we estimate that Texas data centers in 2025 will consume a total of 25 billion gallons (Bgal) of water (77 thousand acre-feet (kAF)) or 0.4% of total water use.¹⁰

By 2030, this demand could increase to 29-161 billion gallons (Bgal) of water consumed¹¹ (88-494 kAF), or 0.5-2.7% of the total annual water use in Texas. This estimate would increase the 804 Bgal (2.5 mAF) needed for the manufacturing and steam electric sectors—the sectors that would capture data center water use and power consumption, respectively—by 4-20% in 2030. These estimates are derived from current and forecasted electricity requirements, as shown below.





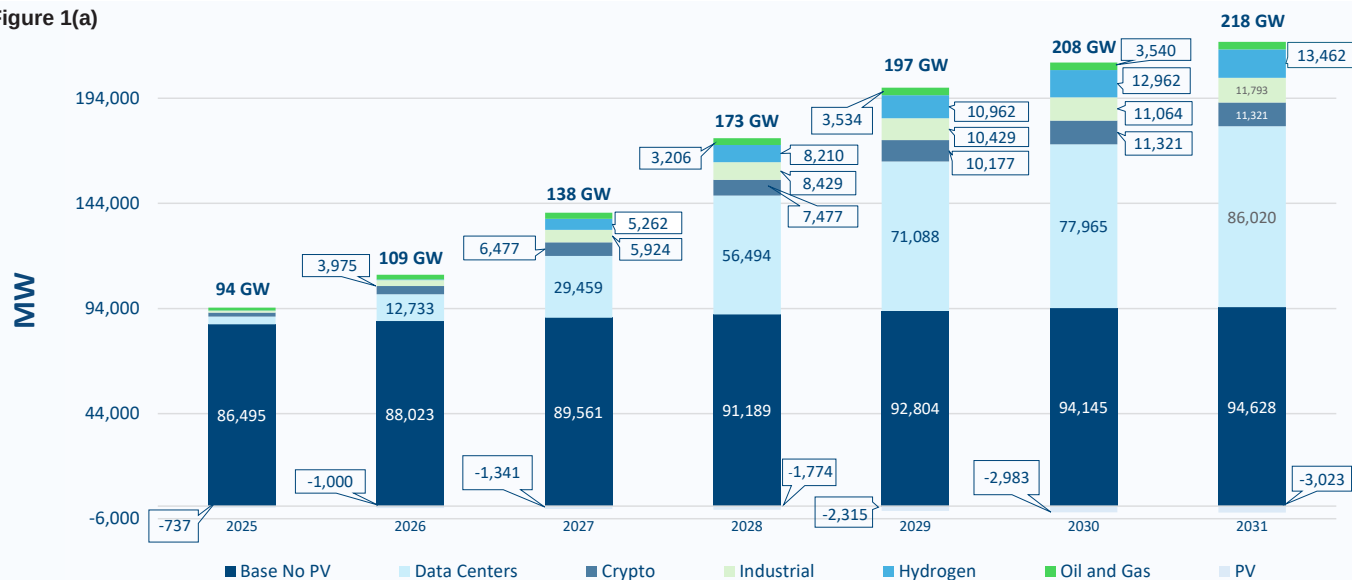
Data Centers and Electricity

According to Baxtel, a marketplace platform for data center research and advisory, Texas currently has 464 data center facilities concentrated in five regions: Dallas Fort Worth (197 facilities), Houston (48), San Antonio (60), Austin (53), and West Texas (59).¹² In September 2025, more than 70 additional sites were under construction with more data centers in planning and development phases.^{13,14} Baxtel estimates the existing data centers require approximately 9,567 megawatts (MW) of power capacity. For context, the Electric Reliability Council of Texas (ERCOT), the state's independent system operator, expects to have 103,105 MW of electric generating capacity available for summer peak load.¹⁵ If running at full steam, existing data centers could require 8% of Texas' current total supply. If these facilities run all day, every day, they would consume 13% of all electricity produced in 2023, the most recent year of data reported to the

Energy Information Administration (via forms 923¹⁶ and 860.¹⁷)

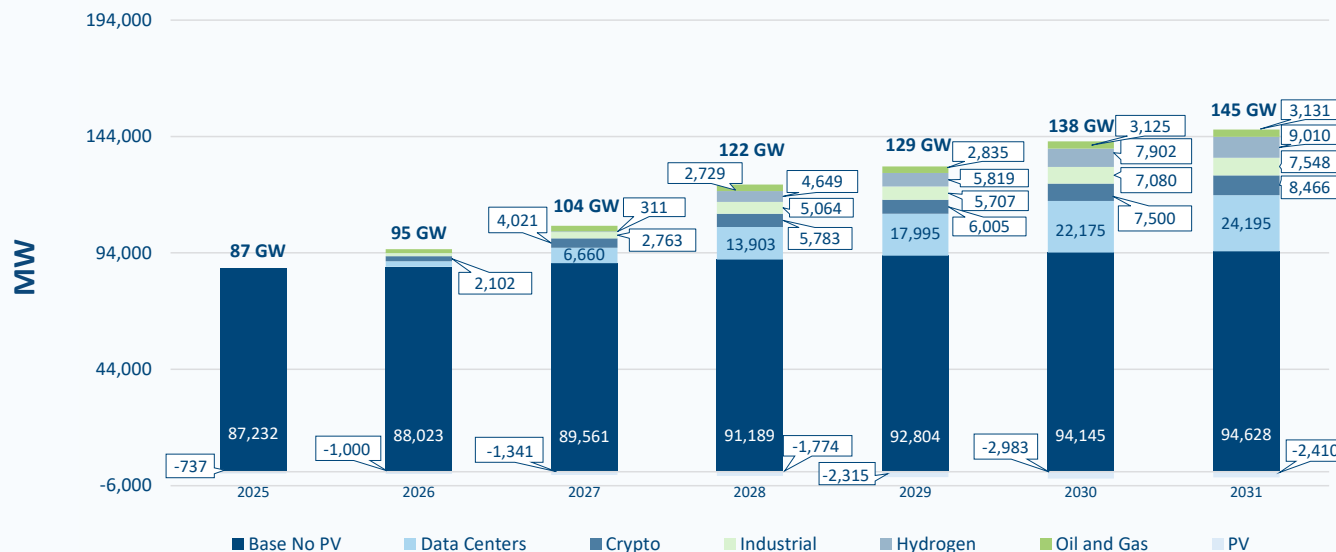
By 2030, Transmission System Providers in ERCOT expect data center electricity demand to increase almost tenfold to 77,965 MW (Figure 1a),¹⁸ an 83% increase over 2025 demands and a worrisome fact since Texas' primary electric grid is already experiencing significant challenges in meeting reliability and resilience needs. ERCOT adjusted this forecast to 22,175 MW (Figure 1b) "based on observation of behavior and characteristics of these loads, including average project delay, load profile by type, and average project realization."¹⁹ While some of the 77,695 MW of load may not be supplied by ERCOT because of the expected delays, data centers are already developing plans for onsite power instead. ERCOT may be able to slow growth on the grid, but other growth may persist.

Figure 1(a)



	Base No PV	Data Centers	Crypto	Industrial	Hydrogen	Oil and Gas	PV
2025	86,495	3,717	1,868	973	5	1,334	-737
2026	88,023	12,733	3,975	2,757	357	2,186	-1,000
2027	89,561	29,459	6,477	5,924	5,262	2,851	-1,341
2028	91,189	56,494	7,477	8,429	8,210	3,206	-1,774
2029	92,804	71,088	10,177	10,429	10,962	3,534	-2,315
2030	94,145	77,965	11,321	11,064	12,962	3,540	-2,983
2031	94,628	86,020	11,321	11,793	13,462	3,606	-3,023

Figure 1(b)



	Base No PV	Data Centers	Crypto	Industrial	Hydrogen	Oil and Gas	PV
2025	87,232	0	0	0	0	0	-737
2026	88,023	2,433	2,102	1,387	5	1,699	-1,000
2027	89,561	6,660	4,021	2,763	311	2,320	-1,341
2028	91,189	13,903	5,783	5,064	4,649	2,729	-1,774
2029	92,804	17,995	6,005	5,707	5,819	2,835	-2,315
2030	94,145	22,175	7,500	7,080	7,902	3,125	-2,983
2031	94,628	24,195	8,466	7,548	9,010	3,131	-2,410

Figure 1 (a) Transmission System Provider Long-term Load Forecast for 2025-2031 provided by ERCOT on April 7, 2025, shows data centers contributing to major electricity sector demand growth.²⁰ (b) ERCOT Adjusted Long-term Load Forecast for 2025-2031 provided by ERCOT on April 8, 2025, shows a 72% reduction in demand in 2030 compared to the TSP forecast in (a) the TSP forecast in (a)

In April 2025, ERCOT approved the Large Load Interconnection Process (LLIP) to manage the interconnection of large loads such as data centers.²¹ ²² When a data center exceeds defined thresholds or is in an area of grid constraint, the data center developer must submit relevant information such as load size and construction timeline. The process includes a Quarterly Stability Assessment (QSA) requirement that large loads be included in ERCOT’s QSA before they are eligible for initial energization. Additionally, large loads must have a load commissioning plan to ensure that new demand does not exceed the grid's capabilities. These rules are intended to ensure there is enough power to supply these large loads prior to their connection to the grid, and compliance with these policies is required.

Data Centers and Water

Data centers often use water for cooling, though some use dry cooling (also called air cooling). Water consumption for cooling is largely evaporation through a chiller or cooling tower. Used water can be reused, but new water withdrawals are needed to replace evaporated water and other system’s water losses. Water demand depends on the type of cooling used, the size, and type of the data center.^{23,24,25} Table 1 shows how water usage varies by cooling type and facility size.



Table 1 Relative Water Usage Effectiveness of Cooling Technologies Used for Data Centers

Cooling Technology	Facility Size		
	Small	Midsize	Large-scale
Direct expansion	Very Low	Very Low	--
Air-cooled chiller	Very Low	Very Low	--
Dry cooler (air-cooled chiller)	--	Very Low	--
Airside economizer (air-cooled chiller)	--	Very Low	--
Airside economizer & adiabatic cooling (air-cooled chiller)	--	--	Very Low
IT Liquid cooling: dry cooler (air-cooled chiller)	--	--	Very Low
Dry cooler with adiabatic assist (air-cooled chiller)	--	--	Low
Airside economizer & adiabatic cooling (water-cooled chiller)	--	--	Low
IT Liquid cooling: dry cooler with adiabatic assist (air-cooled chiller)	--	--	Low
Airside economizer (water-cooled chiller)	--	Medium	--
IT Liquid cooling: waterside economizer (water-cooled chiller)	--	--	Medium
Waterside economizer (water-cooled chiller)	--	High	High
Water-cooled chiller	Very High	High	--

Table 1 is adapted from Lawrence Berkeley National Laboratory “2024 United States Data Center Energy Usage Report”²⁶ where water usage effectiveness (ratio of the total water usage of the data center compared to the electricity demand for the IT equipment) is expressed as Very Low (less than 0.1), Low (0.1-1), Medium (1-2), High (2-3), and Very High (greater than 3). Very High or High represent the greatest water usage effectiveness.



In some cases, air could replace water as cooling fluid as shown in Table 1, but it would require more energy. For example, using an air-cooled rather than water-cooled chiller in a midsize facility would require an average 20% more electricity at the data center,²⁷ and using an air-cooled rather than water-cooled chiller with an airside economizer would require approximately 14% more electricity on average.²⁸ The water consumption of the electricity source should be evaluated before making a decision to switch to air cooling based on water use concerns.

Some data centers use direct evaporative cooling as an adiabatic assist to dry cooling or a cost-effective way to cool air prior to using an airside economizer. Water demand also varies by the type of data center. The water usage effectiveness (ratio of the

total water usage of the data center compared to the electricity demand for the IT equipment) of AI specialized, midsize, and collocated facilities is about double that of their small and hyperscale counterparts.²⁹

Water Consumption in Texas

Lawrence Berkeley National Laboratory estimates data centers in 2023 consumed 95 gallons per megawatt-hour (MWh). At this rate, the 464 data centers in Texas would directly consume 8 Bgal of water (24 kAF) per year for cooling. This volume is 0.1% of the state's estimated annual water use in 2022 reported in the State Water Plan.

Water is also required to cool the power plants that generate electricity used by the data centers. Depending on the location of the power plant and data center, this water could be withdrawn from the same basin or another part of the state. For every MWh generated in 2023, Texas power plants withdrew 12,700 gal (0.039 AF)³⁰ from Texas water supplies and consumed 205 gal (0.0063 AF), according to data reported to the Energy Information Administration (via forms 923³¹ and 860³²). On average, the grid power supplying the 464 data centers in Texas withdraws about 1,067 Bgal (3.3 mAF) and consumes 17 Bgal (53 kAF) per year. This consumption is 0.3% of the estimated annual water use reported in the 2022 State Water Plan.

In total, the indirect and direct consumption is 0.4% of the estimated annual water use reported in the 2022 State Water Plan.

Forecasting 2030 Data Center Water Consumption

In 2030, as power needs may increase by two to ten times, estimated water use for cooling at data centers may increase, as well. Data centers and power supply planners can make significant changes to water use by choosing power supplies wisely. These water and electricity demands pose significant risks to Texas' economy, communities, citizens, and natural resources.

Lawrence Berkeley National Laboratory's "2024 United States Data Center Energy Usage Report" estimates by 2028, data centers would consume 119-127 gal/MWh. Assuming data centers in Texas consume water at this rate in 2030, 77,965 MW of data center demand would consume 81-87 Bgal of water (997 kAF) annually for cooling or 1.4% of Texas' estimated annual water demand. The lower estimate of 22,175 MW of data center demand would consume 23-24 Bgal (71-76 kAF) or 0.4% of Texas' estimated annual water demand.

Scenario 1: Data Centers Using Current Grid Power

Assuming Texas power plants in 2030 will consume water at their current rate, the 22,175 to 77,965 MW of data center electricity demand would consume 40-140 Bgal (122-430 kAF) at power plants, or 0.7-2.3% of total water demand.

Scenario 2: Data Centers Using Updated Grid Power

However, power plants have become more water-efficient over time. Power plants that began operating in the last ten years of the EIA data set (2013-2023) — such as solar, wind, natural gas combined cycle, battery energy storage, or natural gas turbines — withdrew an average 626 gal (2 kAF) and consumed 29 gal (89 AF) from Texas water supplies for every MWh generated.^{33, 34} These additional power plants will power all grid users alongside existing power plants. Under these assumptions, the additional power plants would

cause the entire grid's water demands to shift from 12,700 gal/MWh (0.039 AF/MWh) to 6,300 gal/MWh (0.019 AF/MWh) withdrawn and from 205 gal/MWh (0.0063 AF/MWh) to 120 gal/MWh (0.0004 AF/MWh) consumed. Indirect water use for data centers in 2030 could be significantly lower compared to current indirect water use.

Assuming the additional 22,175 to 77,965 MW of data centers would rely on grid power and these new power plants would employ technologies that use water at a more efficient rate, the marginal additional water consumption would be 6-20 Bgal (17-61 kAF) at power plants or 0.1-0.3% of the total water demand estimated for 2030.

Scenario 3: Data Centers Seeking Onsite Gas

In their assessment of data center growth, ERCOT notes power project delays limiting data center load growth on the grid. While ERCOT estimates 22,175 MW of data center growth may occur on the grid, the remaining data center 55,790 MW in estimated growth may build onsite power plants instead, leading to increased water use across the state. Data centers have already begun exploring direct agreements with power plants rather than relying on the grid. Power providers have reported deals with data centers to provide additional energy supplies directly, including NRG, which plans to build four natural gas combined-cycle (NGCC) power plants totaling 5.4 GW to serve data centers in Texas and Virginia.³⁵ Use of fossil fuels may increase air pollution or emissions. Adding carbon capture to natural gas to mitigate those emissions could double the water demand. Use of carbon capture is not included in this analysis.

Though combined-cycle plants are more water-efficient than other thermoelectric power plants, they still consume water and can add to local

water stress. Texas NGCC plants constructed in the most recent ten years of data (2013-2023), withdrew 2,932 gal/MWh (0.009 AF/MWh) and consumed 109 gal/MWh (0.0003 AF/MWh). These include power plants using dry cooling or water for cooling. NGCC power plants using only water for cooling recirculated through cooling towers would withdraw 297 gal/MWh and consume 244 gal/MWh.

Assuming the additional data centers seek direct contracts with NGCC plants with similar cooling selections to power plants installed in the last decade, they would consume 21-74 Bgal (65-227 kAF) at power plants or 0.4-1.2% of the total water demand estimated for 2030.

Scenario 4: Data Centers Seeking Power Supplies without Cooling Demands

Conversely, if the data centers sought direct contracts for dry-cooled power plants, solar,

solar+storage,³⁶ or other power plants with negligible cooling water demand they could eliminate data center indirect water demand.

Data centers and power supply planners can make significant changes to their water use by choosing their power wisely. In total, current data centers may consume an estimated average 25 Bgal (77 kAF) of water.³⁷ In 2030, this additional water use may increase by a total of 23-161 Bgal (71-494 kAF) or 0.4-2.7% of water demand for that year – increasing total water needs for the entire state of Texas in 2030 by 2-10%. Currently, the manufacturing and steam electric sectors are estimated to use 804 Bgal in 2030 – the sectors that would include data centers and their power use, respectively. Data centers would increase that estimate by 4-20%. The total expected demand varies based on type and size of the data center, the cooling technology at the data center, and the power supply and its cooling needs.

Table 2 Stoplight Table Indicating Water Use of Average Estimated Data Center and Current Grid Power Estimate Compared to Differing Power Supply Choices and Increases in Water and Efficiency.

	1: Current Grid Power	2: New Grid Power	3: Onsite NGCC	4: Onsite Power, No Water for Cooling
Average Data Center Water Use	Base Case			
Energy Efficiency at Data Center				N/A
Water Efficiency at Data Center				Largest Water Savings
Water Efficiency at Data Center (with Energy Penalty)	Increased Water Demand			Largest Water Savings
Water Efficiency in Power Supply				N/A
Water Efficiency at Data Center and in Power Supply				N/A
Water Efficiency at Data Center and Power Supply (with Energy Penalty)				N/A

Green values indicate the largest increases in water savings. The red values indicates an increase in water demand. Color variation in between indicates scale of water decrease. “Current Grid Power” assumes power plants supplied by power already available via the ERCOT grid, calculated from current reported water use and generation estimates. “New Grid Power” assumes new data centers would be supplied by new power plants added to the grid—a combination of solar, wind, energy storage, and gas based on recent trends—and uses water estimates for the most recent decade of reported data to estimate the water use for newly added power plants spurred by data center growth. “Onsite NGCC” assumes power is supplied by NGCC onsite with a cooling tower. “Onsite Power, No Water for Cooling” assumes power is supplied by dry-cooled power plants, solar, solar+storage, or other power plants with negligible cooling water demand. These power plants would not see water demand changes from water efficiency in cooling.

In Table 2, we estimate impacts of water and energy efficiency at data centers or power plants using scenarios 1-3. As the results show, a change in the power supply can have a significant impact on water use. For example, there is a 30% decrease in water use associated with using onsite NGCC compared to current grid power. However, there is a 54% decrease associated with adding new power supply to the grid (a combination of gas, solar, wind, and energy storage). These additional power plants could power data centers directly or power all grid users alongside other power plants. The total water intensity of new power supply and the current grid combined would result in a net 26% decrease in water use with ripple effects of decreased indirect water use across all other users of the Texas grid. Using power sources that do not require water for cooling (scenario 4) would minimize indirect water use for data center operations.

A percent change in energy efficiency is more impactful than an equally sized change in water efficiency at the data center, assuming all other variables are held constant. This difference is

explained by the additional water use associated with power supply. Reducing power use at the data center reduces both onsite and power plant needs for cooling. Conversely, reducing water use at the data center only reduces the data center water consumption. However, a percent change in water efficiency will be more impactful at the data center than in the power supply due to the higher water demand at the data center.

Increasing energy efficiency is more impactful at the data center with all else held constant, but increasing water efficiency is still worthwhile and can have a significant impact. However, managers should be aware of the impact of any corresponding energy penalty. If decreasing water use at the data center (e.g., by pivoting to air cooling) would increase energy use and the power supply is the current Texas grid, total water demand could increase. Conversely, with a water lean power source or one that did not require water for cooling, the water savings would outweigh the water demand associated with increased energy use.

Table 3 Current Water Use Estimates for Data Centers and Power Supplies in Select Regions of Texas

Region	Sites	Data Center Cooling Withdrawal, Bgal (kAF)	Power Plant Cooling Withdrawal, Bgal (kAF)	Power Plant Cooling Consumption, Bgal (kAF)	Percent of Regional Total (%) ³⁸	Total Consumption, Bgal (kAF)
Dallas	197	2 (6)	262 (803)	4 (13)	1%	6 (19)
Houston	48	1 (2)	98 (303)	2 (5)	0.3%	2 (7)
West Texas	59	2 (5)	225 (691)	4 (11)	0.2%	5 (16)
Statewide	464	8 (24)	1,067 (3,273)	17 (53)	0.44%	25 (77)

Estimated using energy data from Baxtel as of April 2025,³⁹ water estimates from Siddik et al. for 2018,⁴⁰ and energy and water data reported to Energy Information Administration forms 923⁴¹ and 860 for 2023.⁴²

Table 3 includes HARC’s estimates for data center water demand by Texas region. Estimated water use for data centers in the Houston and Dallas regions is about 0.3% and 1% of the water demand estimated for Region H and Region D, the regional planning groups for the areas. Estimated water use for data centers in West Texas is about 0.2% of the water demand estimated for Regions A, E, F, and O, the planning group encompassing much of West Texas.

Planning with an Imperfect Tool

Published by the Texas Water Development Board (TWDB), the Texas State Water Plan is the state's primary water planning vehicle. TWDB relies on Water Use Survey Historical Summary Estimates to develop forecasts for water demands and determine water needs. These estimates indicate that water usage for Texas (including reuse) in 2022 was about 5.8 trillion gallons (Tgal) (17.7 million AF).⁴³ The 2022 State Water Plan estimated potential water shortages of around 1.6 Tgal (4.8 million AF) by 2030, and around 2.3 Tgal (6.9 million AF) by 2070.⁴⁴ Because much of the unprecedented growth of data centers has occurred in the years since the water plan was published, their water demands are not fully factored into these numbers.

TWDB's water planning guidance for the State Water Plan notes that market trends for manufacturing, mining, and steam-electric power generation are influenced by many factors and are too unpredictable to forecast. Instead, TWDB relies on reported data in historical water use estimates.⁴⁵ The plan is updated every five years to account for this disparity.

In other words, Texas water plans always look backward, not forward.

That means the 2027 water plan, which is in development now, will be based on 2026 regional water plans that do not include forecasted data center water use. Data centers that began operation in 2025 will not be added to the State Water Plan until 2032.

In some cases, local water plans may include a current or proposed data center. However, it is also possible that local water plans use estimated water supplies based on their forecasted population growth and plan to provide water to data centers with that available supply – essentially borrowing against their future population under the gamble of getting access to more water supplies later.

Communities such as Corpus Christi have taken this approach in the past. Community members and businesses now face possible increased water rates to fund multiple desalination facilities intended to meet the demands from increased industrial



growth.⁴⁶ In public meetings, Corpus Christi leaders stated their planning is hamstrung by limitations on water supply by municipal utilities. Though utilities are required to “provide continuous and adequate water and sewer services to all customers who request it within the designated area,”⁴⁷ there is no clear mechanism at the state level for municipal water suppliers to deny or limit access to certain water users or to require advanced planning information such as location, size, timeline, and expected water demand.⁴⁸ After significant protests and millions invested, the City canceled a contract to build the state's first seawater desalination facility in September 2025.⁴⁹

Analysis Caveats

HARC's estimated impacts reflect forecasts made by ERCOT and estimates of data center water consumption. Should data center growth increase at a less rapid rate, so too will the water impacts. Similarly, this estimate does not consider the nuance of operating times, variety in data center types, sizes, and cooling technologies, or other operation differences that might alter water consumption.

Implications and Recommendations

Better Data Will Lead to Better Planning

Estimates, including those included in this paper, suggest the data center boom will create significant water demand increases. Yet the lack of transparency of data centers makes it difficult to develop precise, verifiable projections. That would require knowing exact demands, technologies used, energy suppliers and other information these companies are not required to publicly disclose.

Moreover, Texas' internal planning authorities are not in sync on data centers. ERCOT anticipates exponential electricity demand growth for the data center industry, but the Texas Water Development Board has not conducted similar forecasting. The State Water Plan process does not provide a mechanism to forecast trends in large loads in the manufacturing, steam electric, and mining sectors.

New data center water demands will get added to future water plans when they are already in operation and appear in the historical water use estimates, meaning water needs estimates could be up to seven years out of date. The State Water Plan is a primary mechanism by which communities get access to funding for new water demands.

If the plan underestimates its unmet water needs, they will not be accounted for in plans for new water supplies, and the state will not be alerted to the need for additional water funding. The water sector is currently underfunded, and the state already has unmet water needs. Moreover, when funding is available, new water projects can take many years to be implemented.

Because of the size and speed of this projected growth, the lack of adequate data and delayed planning processes will hamper communities' ability to plan for these new water demands and ensure they have adequate funding and time to develop new water supplies.

Additionally, location matters. Large demands can disrupt communities and impact water supplies or other industries differently (agriculture, municipal, cattle, etc.). Texas lacks sufficient water use data collection from new industries, and state and local water planning suffers as a result. State and local capacity to make informed water planning decisions is limited by access to this critical information.



Opportunities to Reduce or Offset Water Demands at Data Centers

Various methods could be employed to reduce freshwater demands at data centers or provide new water supplies that could limit their impact on nearby communities:

- **Water Lean Energy Technologies.** Increasing reliance on energy supplies that use less water, like wind, solar, or solar+storage, could significantly decrease the indirect water demand. This change would also ensure that an increase in energy consumption for a water saving strategy such as dry cooling does not increase water demand.
- **Non-Potable Water Supplies at Data Centers or Power Plants.** Data centers and/or power plants may seek alternative water supplies like treated wastewater or brackish water for cooling.
- **Dry Cooling at Data Centers or Power Plants.** Though liquid cooling is more effective⁵⁵, incoming data centers can explore dry cooling technology that uses more electricity but does not consume water onsite. The Sabey Data Center in Round Rock reports using air-cooled units.⁵⁶ Power plants can also employ dry cooling for their operations. Water managers should be aware of the energy penalty and ensure the water savings exceed the water needed for increased energy demand.
- **Adjusted Operating Schedules.** Data centers can adjust their operating times to reduce energy use during periods of peak demand, when the grid may increase its dependence on more water-intensive energy sources.
- **Onsite Energy Efficiency.** Data centers can decrease their water demands by decreasing their energy consumption.
- **Offset Water Use through Other Water-Saving Activities.** Companies can develop or partner to develop projects to save water in the same river basin or aquifer to offset the data center water use, providing a net-water-positive approach. Examples could include

partnering to reduce water leaks or water waste in a nearby municipality.^{57,58}

- **Leverage Infrastructure Investment.** Large data centers may need their own water infrastructure. Infrastructure can be capitalized with the data center as the primary customer and repurposed for public use later as the facility's needs decrease or be constructed large enough to support both needs. For example, an inland brackish water desalination plant constructed for a data center or large-scale long-duration energy storage and excess water could be sold to nearby communities for their water needs.
 - Where shared investment may be more appropriate, communities and data centers can form public-private partnerships to fund new water infrastructure.
 - For an example of a significant need for water savings investment, leaks in Texas water systems account for 186 Bgal (572 kAF).

These and further operating changes could reduce competition for freshwater resources in water-stressed areas of Texas. Because there is no comprehensive survey of data center water demands, however, we cannot report whether current or planned data centers already use these strategies or plan to adopt them. Here, again, better and transparent data will help planners considerably.

Water Planners' Ability to Respond is Limited

Because the state lacks an adequate water supply to meet current needs, reducing water demand at data centers will be a crucial step in ensuring all Texas stakeholders have sufficient water. However, planning for large water demands from data centers is limited at the state and local level.

The state regulates water demands from surface water but not from groundwater or wholesale water sales. There is no "large load" interconnection limitation like that in the electricity sector.

However, the state can play a role in reducing water demands through incentives or requirements for conservation or the use of alternative water supplies. The state could require or incentivize data centers to employ best available cooling technology or air cooling, which would limit the water use of data centers.

Cities are limited in their authority to reject an entity submitting for a water connection in their geographic service area due to their requirement to “provide continuous and adequate water and sewer services to all customers who request it within the designated area.” They can, however, leverage water conservation ordinances to encourage reduced water use. Cities may consider incentivizing or requiring water-lean technologies for large load facilities or the use of alternative water supplies, such as treated wastewater effluent. Communities can also leverage investment or enter public-private partnerships with data centers to fund new water supplies that can benefit both the data center and water customers. Without this investment, the increased infrastructure needed could drive up costs that, if not paid by data centers now, would get passed across entire service areas through increased water rates.

Groundwater conservation districts (GCDs) across the state have different rules and may limit total volume in certain districts or only have authority to limit well spacing or pumping rates. These varied sets of controls hinder a district’s ability to adequately protect groundwater supplies from being depleted by large demands. Brackish water may or may not be regulated by the local GCD, meaning it may be a viable source of unregulated water for data centers in water-stressed areas. However, water planners should be careful to ensure that depletion of brackish water supplies does not impact fresh groundwater or streamflow as aquifer systems can be connected.

The convergence of high-water consumption, limited transparency, and already-stressed supplies presents a critical and growing challenge for Texas. Without coordinated action, the state risks compounding its water scarcity issues and increasing water costs for communities⁵⁹ that



already struggle to access and afford water, face the risk of running out of water, or are forced to consider environmentally risky decisions for new water supplies.

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- ¹⁰ Total water use is 5.8 trillion gallons or 17.7 million acre-feet.
- ¹¹ Withdrawal means water pulled from a water source. Consumption means water that is used and not returned to the water source. Water that is withdrawn but not consumed would be discharged or returned to the water source.
- ¹² <https://baxtel.com/data-center/texas>
- ¹³ <https://baxtel.com/data-center/texas>
- ¹⁴ Other entities have varying numbers on data centers in Texas. Some of this variation is due to how data centers are tracked (some only qualify larger data centers, some by power), but some is due to lack of transparency and streamlined data on data centers. For example, the Texas Comptroller lists 279 data centers in the state and Data Center Map (a marketplace platform for data center services) lists about 340.
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- ²⁷ The average power usage effectiveness (PUE) or total power usage of the facility divided by the electricity use for IT equipment for a water-cooled chiller is ~1.5. The average PUE of an air-cooled chiller is ~1.8, a 20% increase.
- ²⁸ The average PUE of an airside economizer with water-cooled chiller is ~1.4. The average PUE of an airside economizer with air-cooled chiller is ~1.6, a 14% increase.
- ²⁹ <https://eta.lbl.gov/publications/2024-lbnl-data-center-energy-usage-report>
- ³⁰ Water is withdrawn to cool steam used for generating electricity. Much of that water may be returned to the environment warmed by the power plant, particularly in cases where the water only flows from the water source through the power plant once (once-through cooling). However, to reduce withdrawals, power plants may recirculate water for cooling using cooling towers or ponds (recirculating cooling). Because the same pool of water is used as both the source

and the return supply, it will be warmer and require evaporation to cool to a useful temperature. This evaporated water is consumed.

³¹ <https://www.eia.gov/electricity/data/eia923/>

³² <https://www.eia.gov/electricity/data/eia860/>

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³⁶ Certain types of energy storage do consume water for their preparation, e.g., saltwater storage consumes water in filling the battery with salt water.

³⁷ The Texas Water Development Board estimates that the entire livestock industry consumes 108 Bgal (332 kAF) per <https://texasstatewaterplan.org/>

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